Adapting Differentiable CFE into NeuralHydrology to Train CFE Parameters for Use in NextGen

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Background

- NextGen requires parameter estimation but it is challenging especially at locations with insufficient observations to constrain the model.
- CFE is a simplified conceptual model functionally equivalent to the NWM [3] and needs calibrated parameters.
- Differentiable ML has shown potential, here's how it works [4]:

$$\widehat{Q} = g^{\theta}(u, x, A), \tag{1}$$

here, g is the CFE model that depends on parameters $\theta = f^{W}(u, x, A)$, and f is a LSTM with weights found by minimize some loss function $W = \operatorname{argmin} L(Q, \widehat{Q})$.

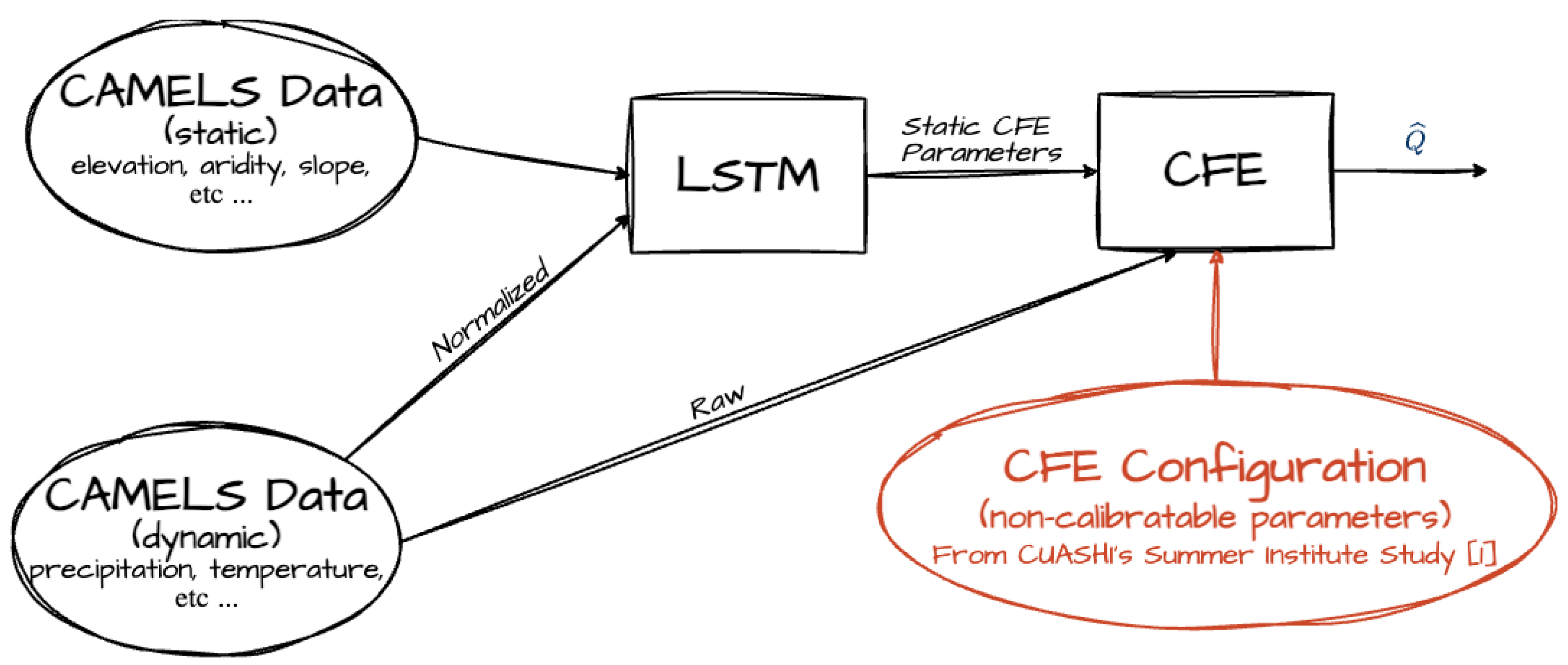

Figure 1. Diagram of data flow for a forward pass in NH-dCFE. The item in orange contain CFE configuration and calibrated parameters from a previous study from the Consortium of Universities for the Advancement of Hydrological Science [1], which involved creating new structure in NeuralHydrology (NH) to integrate. CFE is roughly 3 - 4 times more complex than SHM or HBV.

We investigated parameter regionalization for NextGen's CFE model by embedding a differentiable version of CFE (dCFE) into the NeuralHydrology (NH-dCFE) platform [2] for training and extracting the trained LSTM to use in CFE parameter regionalization across CONUS.

Validation to CFE

Validating our implementation of NH-dCFE to original author code's streamflow related outputs.

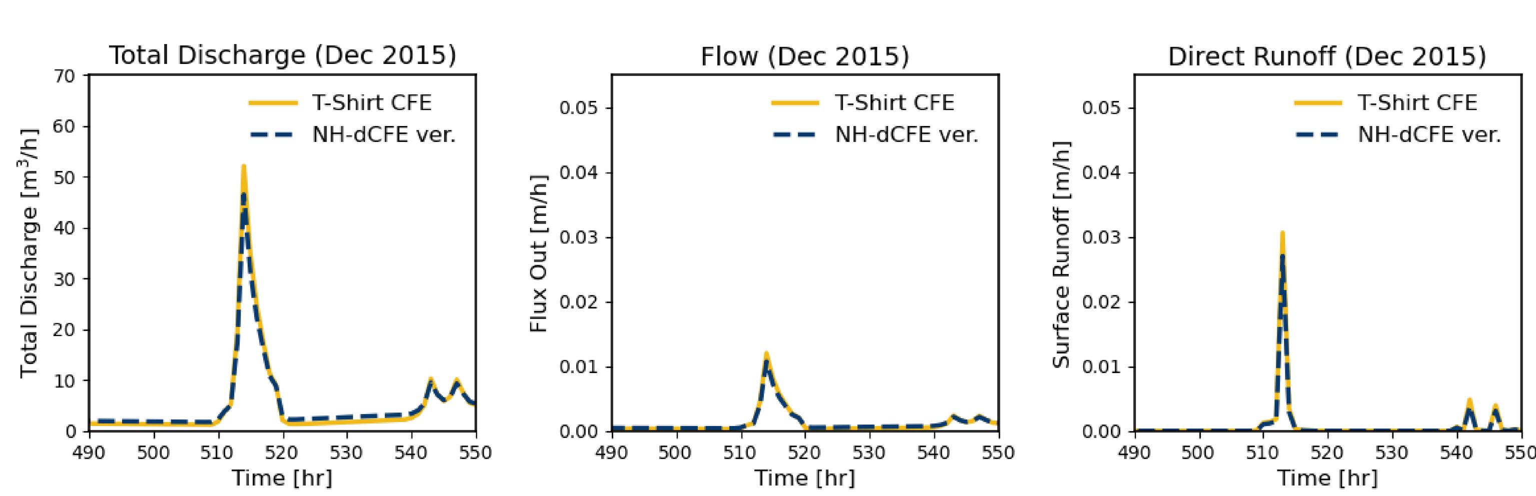

Figure 2. Time-series snapshot of validating NH-dCFE to original CFE [3].

Validation to Previous Calibration Study

Validating our implementation of NH-dCFE a previous study's streamflow outputs.

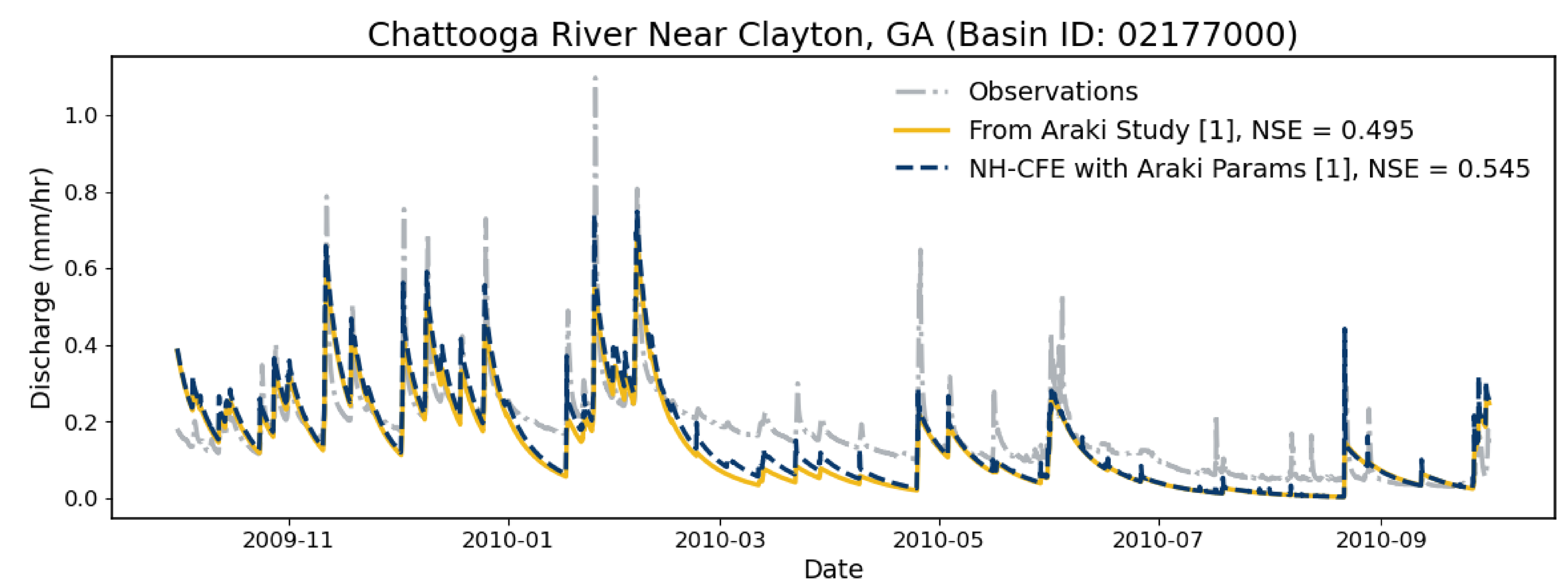

Figure 3. Plot shows that our implementation using Araki calibrated parameters yield similar streamflow to the Araki study [1].

Training & Performance

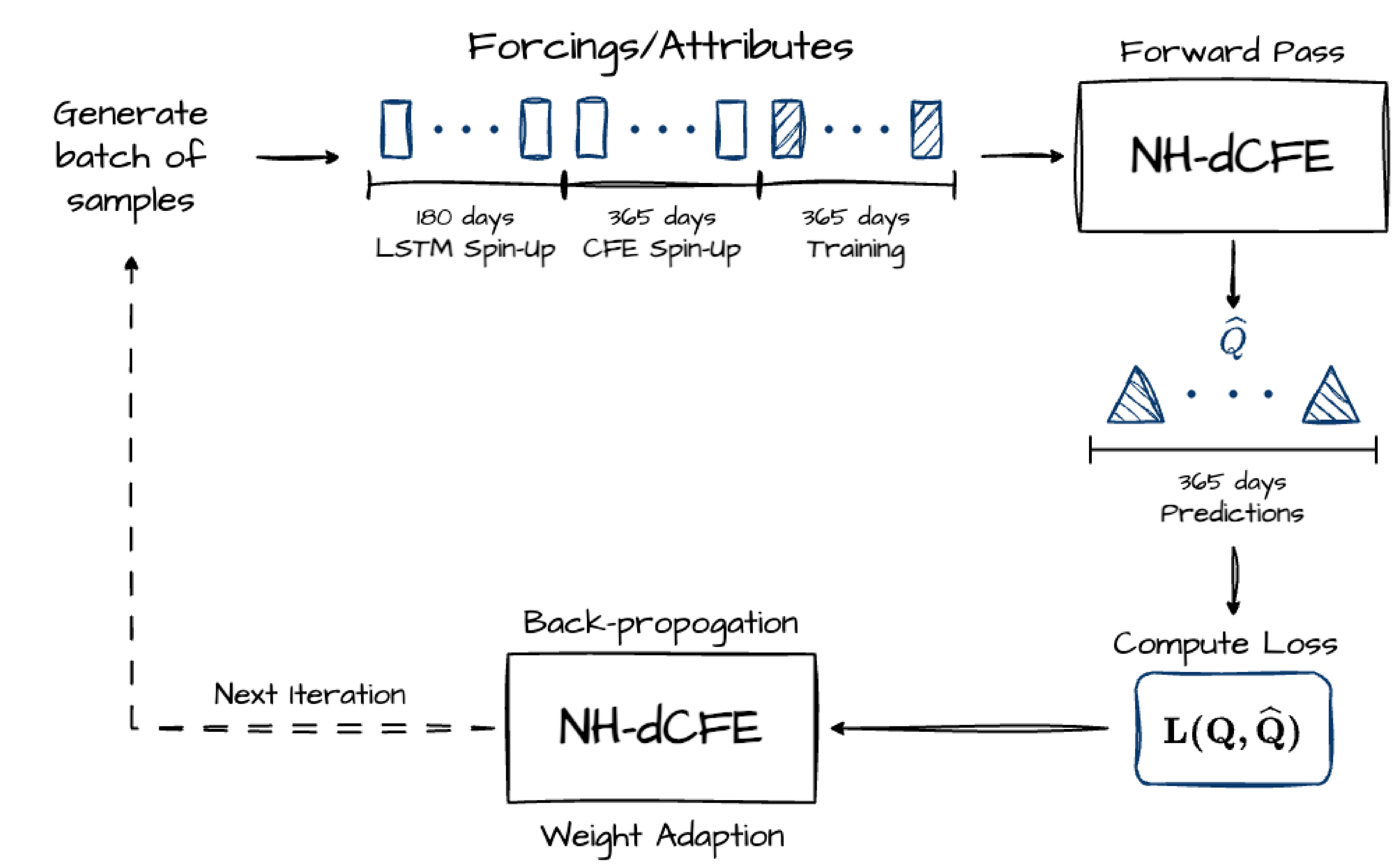

Figure 4. Diagram of each iteration of the training process. Training set-up: 1/2 years LSTM spin-up, 1 years CFE spin-up after, and then 1 year daily outputs for back-propagation.

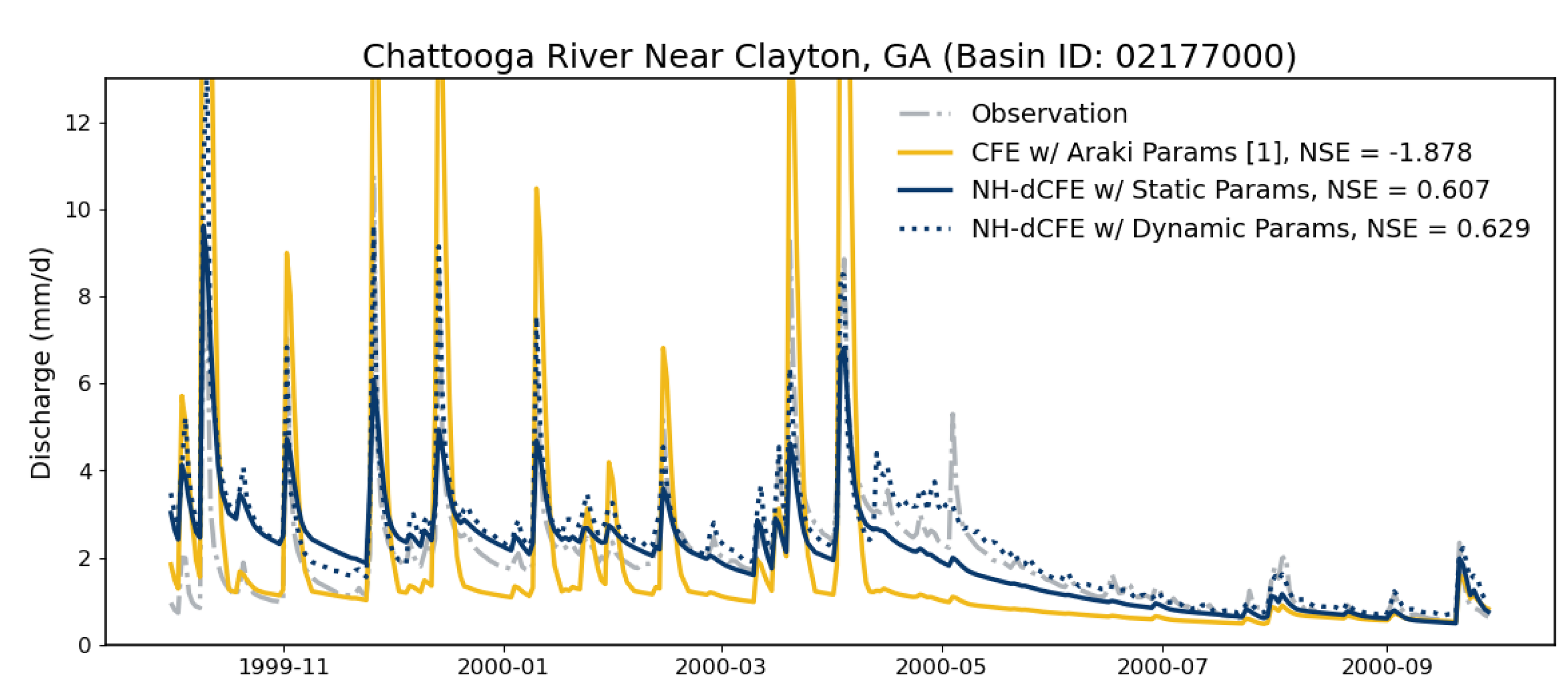

Figure 5. Plot shows that while the NH-dCFE with dynamic parameters, as traditional to differentiable modeling, does do better in this test case compared to static parameters but the difference is not drastic.

Resulting CFE Parameters

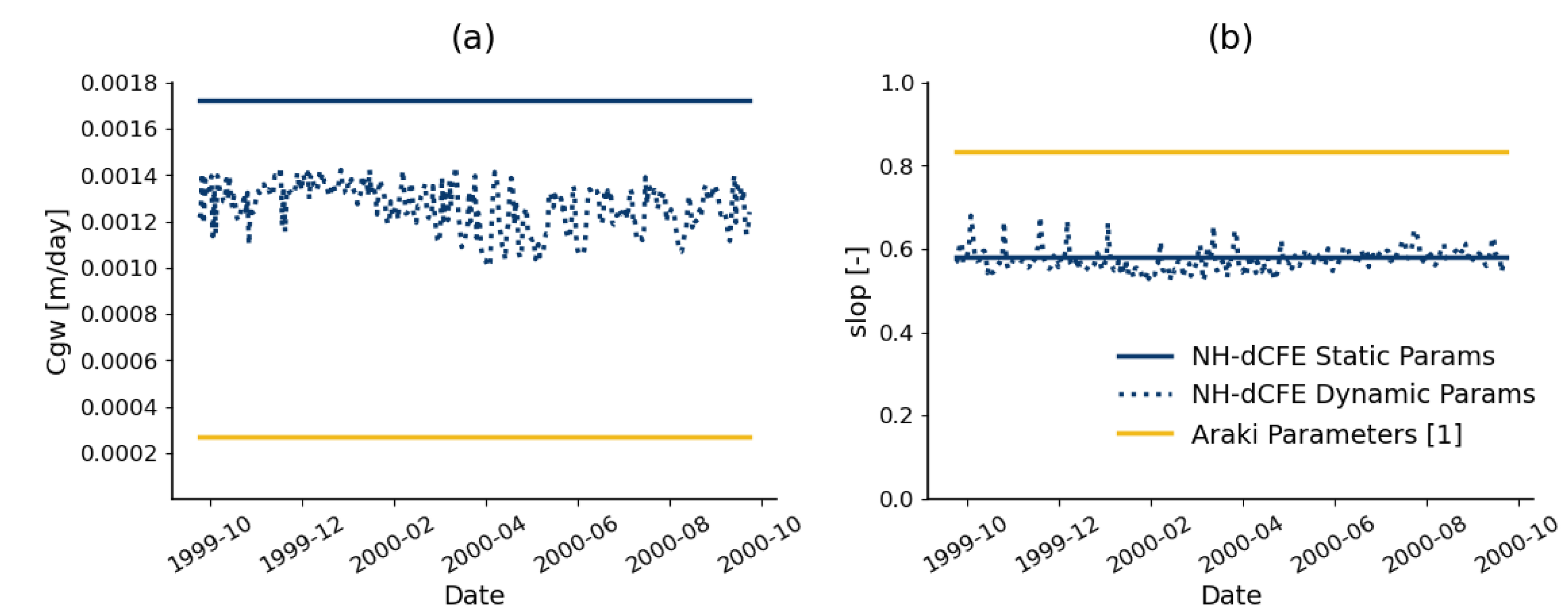

Figure 6. Both plots show time-series of optimal parameter values from the Araki study and from two NH-dCFE trained to generate static vs. dynamic parameters for Chattooga River Near Clayton, GA. (a) shows a CFE parameter associated with groundwater, and (b) shows one associated with slope. NH-dCFE trained to predict dynamic vs. static parameters seem to arrive at similar conclusion for (b) but not for (a).

Summary

Successes:

- Implemented dCFE into NeuralHydrology using some built-in functionality.
- Created tools for exporting CFE parameters from trained model in NeuralHydrology to NextGen.
- Created tools to examine internal states.

Challenges:

- Long training times for hourly model.
- Interpretability and identifiability.
- Scalability.
- Lack of snow module limits number of basins that can be used for training.

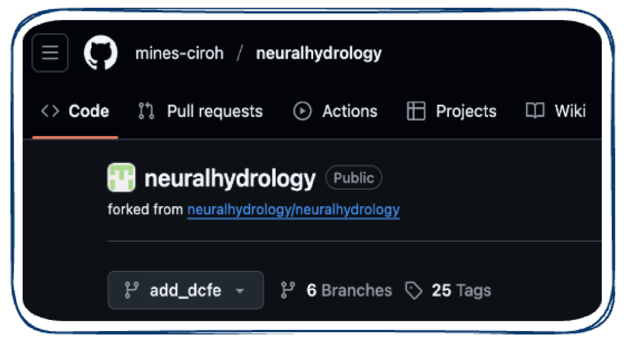

Figure 7. Working repository on Mine's fork of NeuralHydrology.

Future Work

- Ablation study to test the sensitivity and identifiability of each CFE parameter.
- Refine code and merge back to official NeuralHydrology.
- Resolve computational issues to enable hourly training.
- Include snow module.

References

[1] R. Araki, S. A. Bhuiyan, L. Cunha, T. Bindas, J. Rapp, and J. M. Frame. Theme-1 nextgen-aridity: Cfe config & outputs. https://doi.org/10.4211/hs.f7d6db8f8677402d808531924bbcf60c, 2025.

[2] F. Kratzert, M. Gauch, G. Nearing, and D. Klotz. Neuralhydrology — a python library for deep learning research in hydrology. *Journal of Open Source Software*, 7(71):4050, 2022.

[3] F. Ogden. GitHub - NOAA-OWP/cfe: CFE is a conceptual rainfall-runoff model with an implementation of the Basic Model Interface. — github.com. https://github.com/NOAA-OWP/cfe. [Accessed 22-05-2025].

[4] Y. Song, W. J. M. Knoben, M. P. Clark, D. Feng, K. Lawson, K. Sawadekar, and C. Shen. When ancient numerical demons meet physics-informed machine learning: adjoint-based gradients for implicit differentiable modeling. *Hydrology and Earth System Sciences*, 28(13):3051–3077, 2024.